Aryan G Bhojwani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.25
Ratna Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	97.50%
St. Paul's High School	Class X (Board of Secondary Education, Telangana)	2022	9.80%

PROJECTS

Multithreaded Reverse Proxy and Load Balancer | *C++, POSIX Sockets, HTTP, Multithreading*

- Built a multithreaded reverse proxy from scratch in C++ using **POSIX sockets**, implementing **TCP connection management**, HTTP request forwarding, and response relaying between clients and backend servers.
- Developed a custom **HTTP/1.1 parser** to process request lines and headers, enabling request inspection, routing, and modular handling without external networking libraries.
- Implemented **round-robin load balancing** across multiple backend servers with configurable backend selection, connection management, and automatic request forwarding.
- Designed a modular architecture separating networking, HTTP parsing, routing, proxy logic, and backend communication.

Computational Number Theory Library | *C++, Cmake*

- Developed a modular **C++** library implementing advanced number theory algorithms, including **Miller-Rabin** primality testing, **Pollard's Rho** factorization, **Baby-Step Giant-Step**, and the **Chinese Remainder Theorem**.
- Implemented efficient prime generation and arithmetic utilities such as the **Sieve of Eratosthenes**, **Linear Sieve**, **Segmented Sieve**, **Euler's Totient**, **Möbius Function**, and divisor functions with optimal asymptotic complexities.
- Built reusable modular arithmetic components including binary modular exponentiation, modular inverse, and the Extended Euclidean Algorithm, leveraging 128-bit arithmetic for robust 64-bit computations.
- Designed a comprehensive testing and benchmarking framework using **CMake** and std::chrono to validate correctness and evaluate the runtime performance of all implemented algorithms.

Mutual Fund Portfolio Optimizer | *Python, Streamlit, SciPy, Matplotlib*

- Developed a portfolio optimization application implementing **Markowitz Mean-Variance Optimization** to construct Minimum Variance and Maximum Sharpe Ratio portfolios
- Built a portfolio analytics engine to compute annualized return, volatility, **Sharpe ratio**, CAGR, covariance, and correlation matrices from historical mutual fund NAV data.
- Implemented an **Efficient Frontier** generator by solving a sequence of constrained optimization problems (SciPy SLSQP) across target returns and visualized risk-return trade-offs.
- Designed an interactive **Streamlit dashboard** for mutual fund analysis featuring portfolio optimization, allocation charts, and correlation heatmaps.

Interactive Graham Scan Visualizer | *C++, Raylib*

- Completed under the guidance of Prof. J Balasubramaniam in a group of 2 members as part of Data Strucutres and Applications Lab course
- Developed an **interactive C++ application using Raylib** to visualize the Graham scan on randomly generated 2D point sets.
- Implemented subroutines for orientation tests, cross products, and polar-angle sorting to compute the convex hull efficiently.
- Animated each stage of the algorithm, illustrating point selection, stack operations, and incremental convex hull construction.

ACHIEVEMENTS

IICPC Codefest 2026 ranked **2784** among 13000+ participants

Achieved title of **Specialist** on Codeforces (max rating: **1480**)

SKILLS

Languages: C, C++, JavaScript, Python, Go

Libraries and Frameworks: React.js, Express.js, Gin, Pandas, Numpy, Matplotlib

Databases: PostgreSQL, MongoDB

Dev Tools: npm, Postman, Git

RELEVANT COURSES

Computer Science: Data Structures and Applications, Data Structures and Applications Lab, Algorithms, Theory of Computation, Compilers I, Database Management Systems I, Introduction to Programming

POSITIONS OF RESPONSIBILITY

Core Member (Software Domain) | Tinkerer's Lab

- Conducted a technical session covering version control and collaborative software development practices.
- Successfully completed the Software Summer School, receiving hands-on training in React.js, HTML, CSS and Git

Web Coordinator, Inter IIT Sports Meet 2025

- Built pages and reusable UI elements for the Inter IIT Sports Meet 2025 website
- Built components for the mobile app
- Learned about working with multiple people and using collaborative tools efficiently such as Git and Github